

# Everything is *Connected!*

Space, Time, and Graphs

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**“Space-time-causation, or name-and-form,  
is what is called Maya.”**

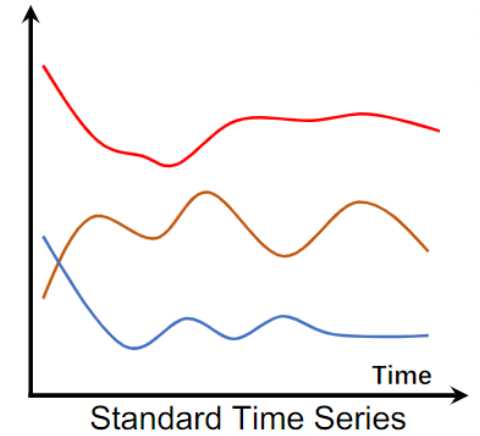
— Swami Vivekananda



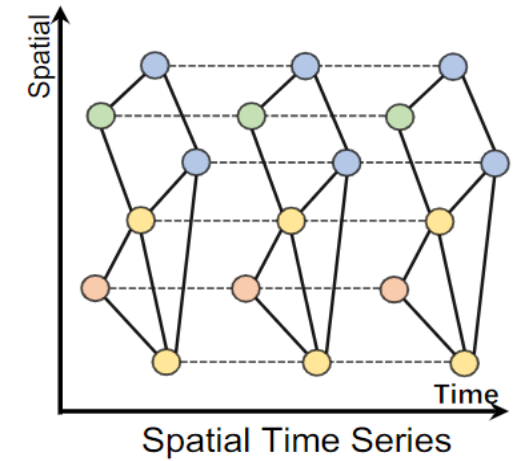
# Time and Space



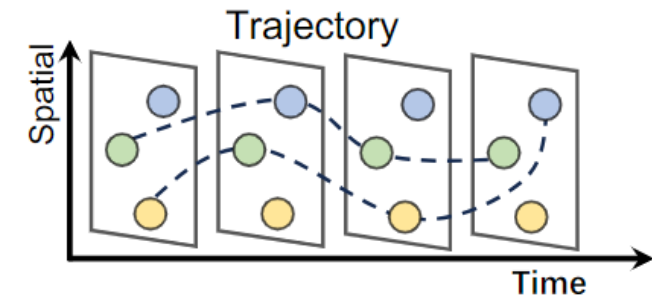
**Time series** is a set of observations, each one being recorded at a specific time.



A **spatial time series** is a type of data that combines spatial and temporal dimensions, capturing how measurements or observations vary across different locations over time.



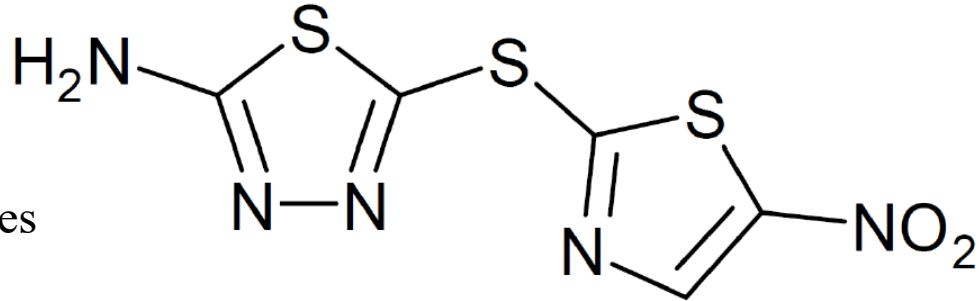
**Spatio-temporal data** is vital for analyzing phenomena in both space and time, providing a dynamic perspective on complex system interactions.



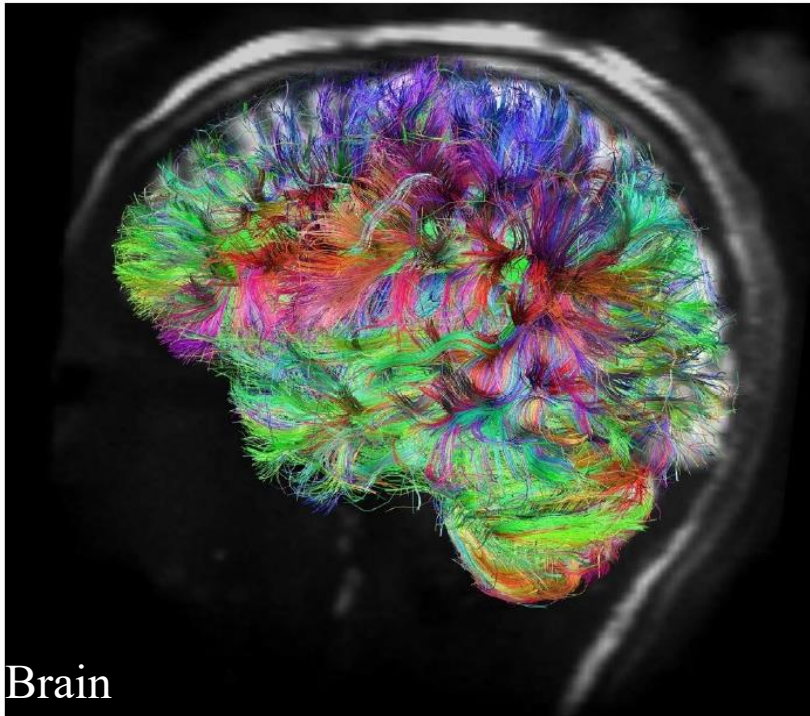
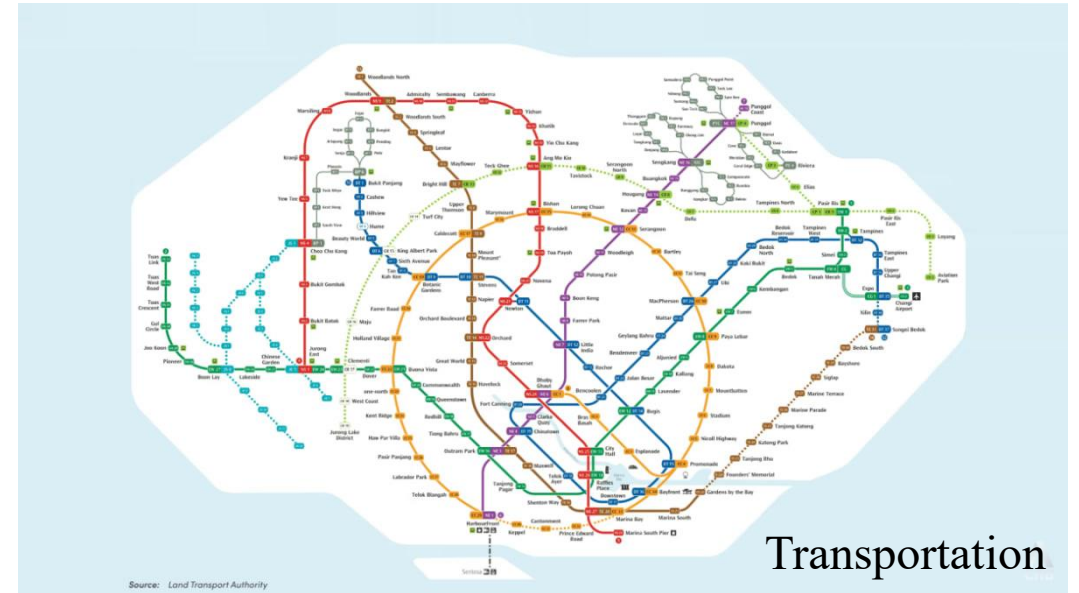
**Forecasting** is estimating how the sequence of observations will continue into the future.

# Graphs are *everywhere*!

In many ways, **graphs** are the **main** modality of data we receive from **nature**.



Molecules

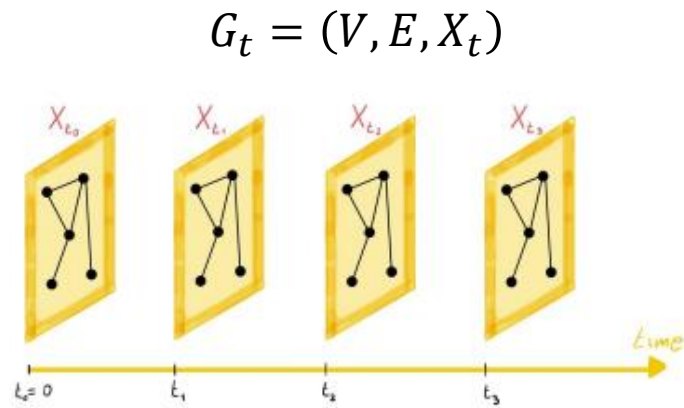


Brain



# Space-Time Graph

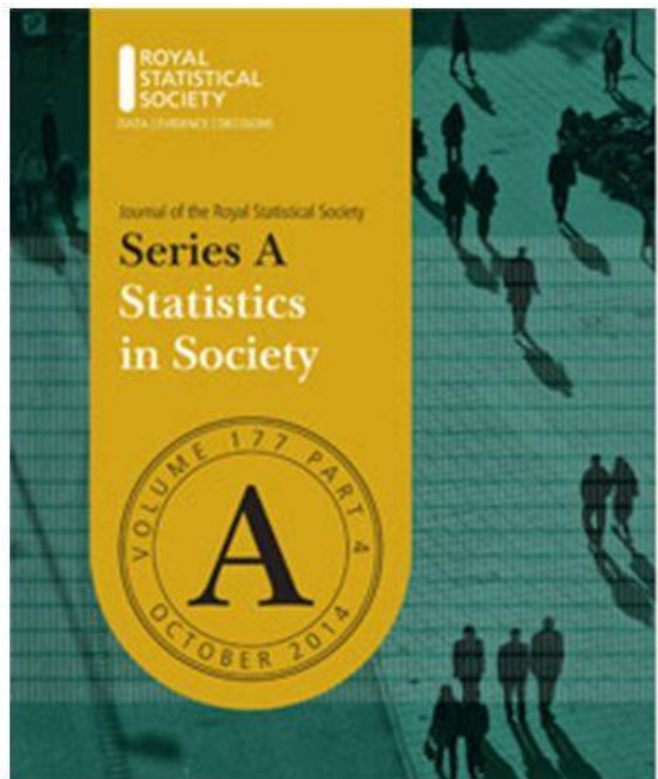
- **Graph  $G(V, E, W)$ :**  $V$  = nodes,  $E$  = edges (link node pairs),  $W$  = weights associated with edges.
- A **dynamic graph** is a type of graph whose structure (nodes and edges) and/or associated properties (attributes of nodes,  $X_t$ ) change over time.



*Spatio-Temporal Graph*

- Topology is fixed, but features change over time
- (Usually) observed at regular intervals
- Examples: Air pollution level

# Today's Talk



PAPER

## E-STGCN: Extreme Spatiotemporal Graph Convolutional Networks for Air Quality Forecasting

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# Air Quality: A Global Concern

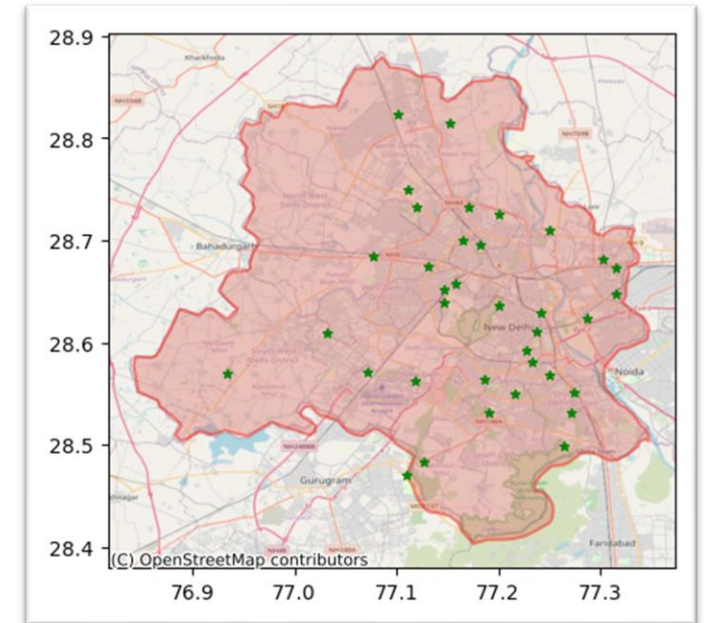


- Emission of particulate matter and harmful gases in the air from industrialization degrades air quality levels leading to harmful health hazards.
- WHO has set National Ambient Air Quality Standards (NAAQS) for major air pollutants to protect the public health and environment.
- Despite several efforts, Delhi (Capital of India) remains the most polluted capital city in the world as of early 2024.

How can we accurately forecast AQIs for Delhi?

# Air Quality Data of Delhi

- In efforts to monitor and reduce the air pollution levels, the Central Pollution Control Board (CPCB) in Delhi has installed a network of forty ground-based sensors at different strategic locations.
- The sensors collect near real-time (hourly) data on the concentration levels of major air pollutants, e.g.,  $\text{PM}_{10}$ ,  $\text{PM}_{2.5}$ ,  $\text{NO}_2$ .
- This timely information empowers the pollution control management system, facilitating informed decision-making to reduce pollution levels.



# Data Characteristics

We collected daily pollutant concentration levels of  $\text{PM}_{2.5}$ ,  $\text{PM}_{10}$ ,  $\text{NO}_2$  monitored by the 40 sensors of Delhi during 1<sup>st</sup> January, 2019 till 31<sup>st</sup> December, 2023.

- Nonlinear
- Non-stationary
- Long-term dependency
- Seasonality
- Extreme events

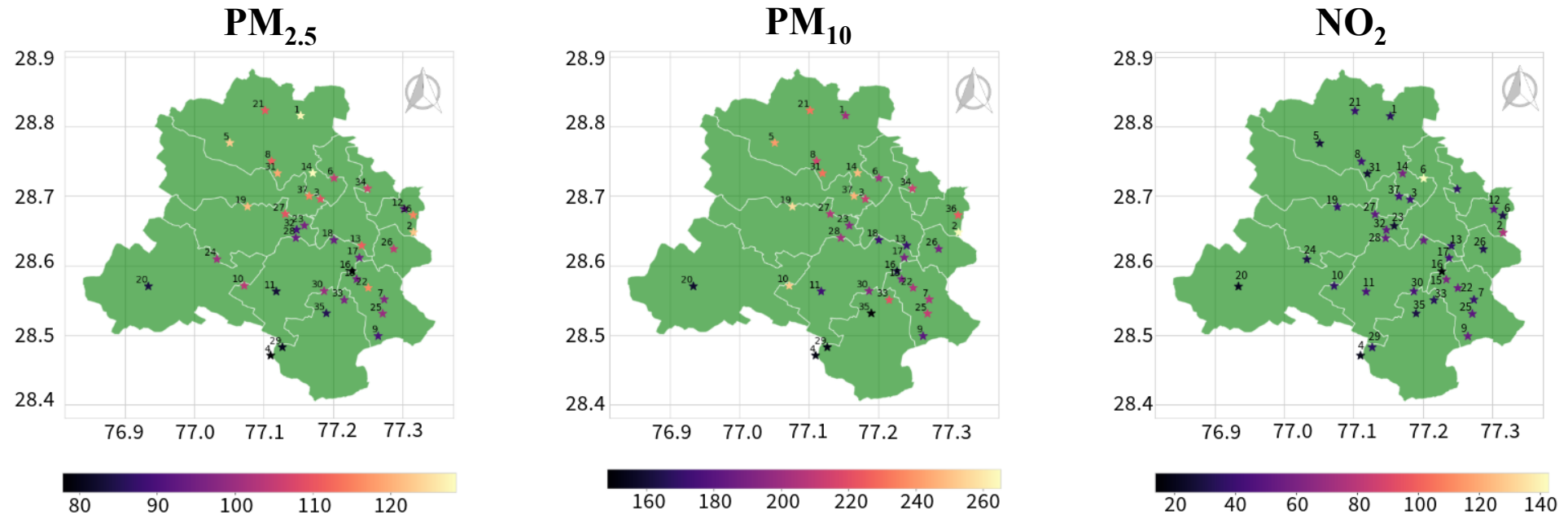


Fig: Average pollutant concentration recorded in Delhi during 2019 – 2022.

NAAQS Thresholds for  $\text{PM}_{2.5}$  is  $60 \mu\text{g}/\text{m}^3$ ,  $\text{PM}_{10}$  is  $100 \mu\text{g}/\text{m}^3$ , and  $\text{NO}_2$  is  $80 \mu\text{g}/\text{m}^3$ .

$\mu\text{g}/\text{m}^3$  denotes micrograms per cubic meter air

# Extreme Value Theory

- In classical statistics: focus on average behavior of stochastic process (Central Limit Theorem)
- In extreme value theory: focus on extreme and rare events (Fisher-Tippett Theorem)
- Block Maxima Method

$$M_n = \max\{X_1, \dots, X_n\}$$

$M_n$  follows a Generalized Extreme Value distribution

- Peak over Threshold (PoT)

$$\{X_i - \tau \mid X_i > \tau\}$$

very large threshold  $\tau$  follow a Generalized Pareto distribution (GPD)

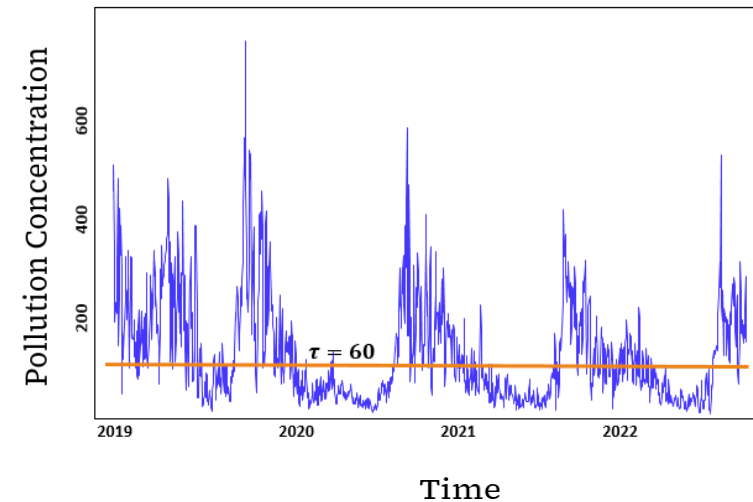


Fig: PoT method on  $PM_{2.5}$  level in Alipur, Delhi

# Extreme Values in AQI Data

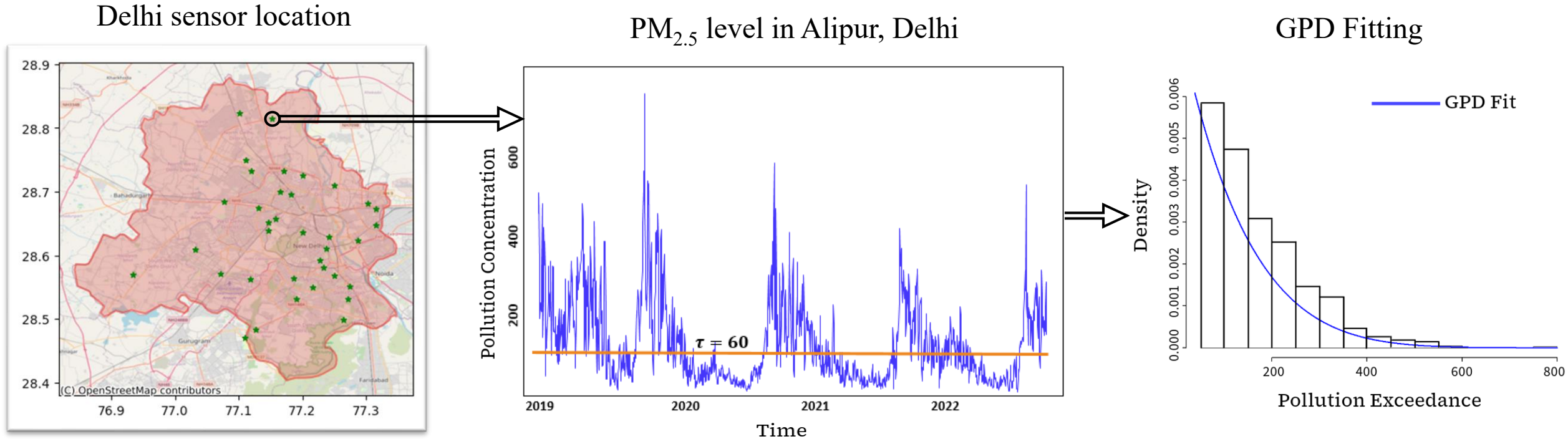


Fig: PoT analysis of PM<sub>2.5</sub> concentration level recorded daily during 2019 – 2022 at Alipur, Delhi

- In Delhi, the air pollution level increases with the onset of winter due to the land-locked geographical location and low temperature restricting the dispersion of the pollutants (**Extreme measurements arise**).

# Proposed Solution: E-STGCN

- How to forecast air pollution levels in a spatiotemporal setup in the presence of extreme values?
- We utilize **PoT approach** from EVT to provide prior information to the Graph Neural Network about exceedances of air pollutant concentration levels.
- **E-STGCN** better optimizes the model parameters to produce accurate forecasts in the presence of multiple peaks (**extreme values**) in the dataset.

# Graph Neural Networks (GNNs)

- GNNs are an optimizable transformation on all attributes of the graph that preserves graph symmetries.

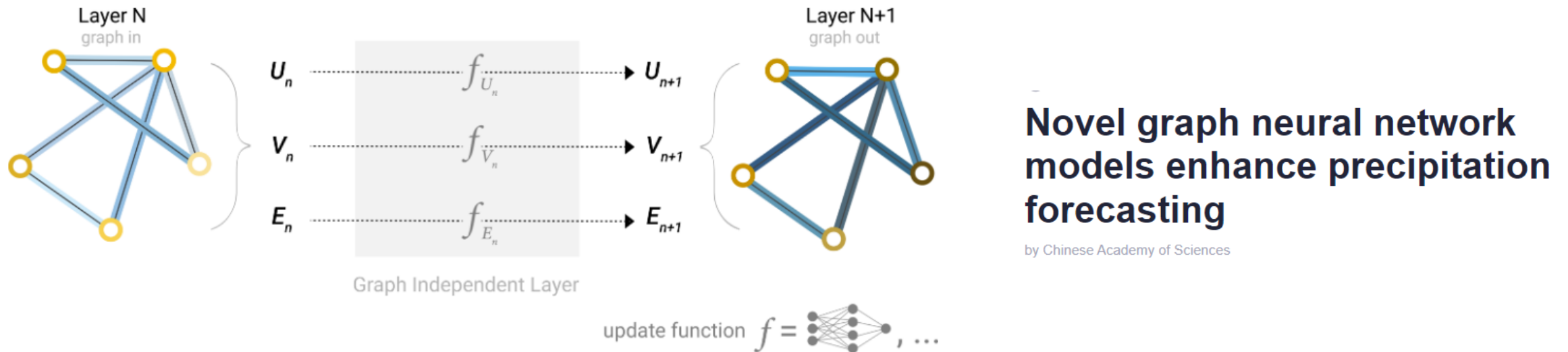


Fig: A single layer of a simple GNN.

- GNNs adopt a **graph-in, graph-out** architecture i.e., it accept a graph as input, with information loaded into its nodes ( $V$ ), edges ( $E$ ), global-context ( $U$ ), and progressively transform these embeddings using distinct MLPs, without changing the connectivity of the input graph.

# Prediction task using GNNs

- For the prediction task we apply a regressor or classifier on each of the learned node representations.
- **GCNs** architectures apply convolutional operations for information propagation between nodes in a graph.

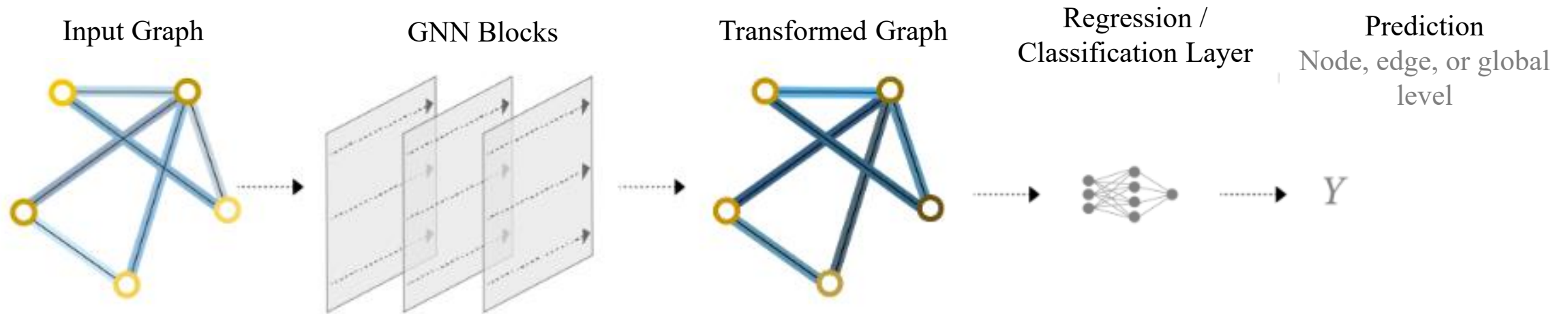


Fig: An end-to-end prediction-task using GNN.

# E-STGCN Framework

- **Input:** Pollutant concentrations across  $N$  sensors at timestamp  $t$ ,  $\mathbf{X}_t = \{X_t^1, \dots, X_t^N\} \in \mathbb{R}^N$ ;  $t = 1, \dots, T$ .
- **Output:**  $q$  ( $\geq 1$ ) steps ahead forecast  $\hat{\mathbf{X}}_{T+1}, \dots, \hat{\mathbf{X}}_{T+q}$  based on  $T$  historical observations.
- **Procedure:**

- Design a spatiotemporal graph with nodes (sensor locations) and edges (interconnections).
- The input can be represented as  $\mathcal{G} = \{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_T\}$  with  $\mathcal{G}_t = \{\mathbf{X}_t, A\}$  being the graph at time  $t$  and  $A \in \mathbb{R}^{N \times N}$  being the static adjacency matrix.
- Compute  $A = [a_{ij}]_{N \times N}$  using the **weighted Haversine distance** as

$$d_{ij} = 2R \sin^{-1} \left[ \sqrt{\sin^2 \left( \frac{\phi^i - \phi^j}{2} \right) + \cos(\lambda^i) \cos(\lambda^j) \sin^2 \left( \frac{\lambda^i - \lambda^j}{2} \right)} \right]$$
$$a_{ij} = \exp \left( -\frac{d_{ij}^2}{\sigma^2} \right), \text{ when } i \neq j, \exp \left( -\frac{d_{ij}^2}{\sigma^2} \right) \geq \epsilon$$

where  $\phi^i, \lambda^i$  is the latitude, longitude of sensor  $i$ , with  $\sigma^2, \epsilon$  controlling the sparsity of  $A$ , and  $R$  represents Earth's radius.

# E-STGCN: Spatial Module

- Encode the structural and feature-based aspects of the graph using GCNs ( $K$  layers) through neural message passing mechanism for node  $i$  as follows:

$$h_t^{i,(0)} = X_t^i$$
$$h_t^{i,(k)} = f_t^{(k)} \left( W_t^{(k)} \frac{\sum_{j \in \mathcal{N}(i)} h_t^{j,(k-1)}}{|\mathcal{N}(i)|} + B_t^{(k)} h_t^{i,(k-1)} \right); k = 1, 2, \dots, K$$

- Generate the spatiotemporal embedding based on the GCNs representations using an MLP as:

$$Z_t^i = \text{Dense} \left( h_t^{i,(K)} \right)$$

# E-STGCN: Temporal and EVT Modules

- Provide the spatiotemporal embedding  $\mathbf{Z}_t = \{Z_t^1, \dots, Z_t^N\}$  to the **Temporal Block: LSTM layers and a Dense layer** to generate the desired forecasts.
- In the EVT module perform PoT approach on the pollutant data.
- Hence, the conditional GPD for the  $i^{th}$  monitoring station at time  $t$  is:

$$P[X_t^i - \tau \leq x_t^i \mid X_t^i \geq \tau] = \begin{cases} 1 - \left(1 + \frac{\xi^i x_t^i}{\alpha^i}\right)^{-1/\xi^i}, & \xi^i \neq 0 \\ 1 - \exp\left(-\frac{x_t^i}{\alpha^i}\right), & \xi^i = 0 \end{cases}$$

where  $\xi^i (\in \mathbb{R})$  is the shape parameter and  $\alpha^i (> 0)$  is the scale parameter.

# E-STGCN: Optimization

- Optimize the network using the following loss function (based on the data loss and the PoT-based loss):

$$\text{Loss}(Z_t^i, \hat{Z}_t^i) = \begin{cases} \text{MSE}(Z_t^i, \hat{Z}_t^i), & \hat{Z}_t^i \leq \tau \\ \beta_1 \text{MSE}(Z_t^i, \hat{Z}_t^i) + \beta_2 \text{PoTL}(\hat{Z}_t^i), & \hat{Z}_t^i > \tau \end{cases}$$

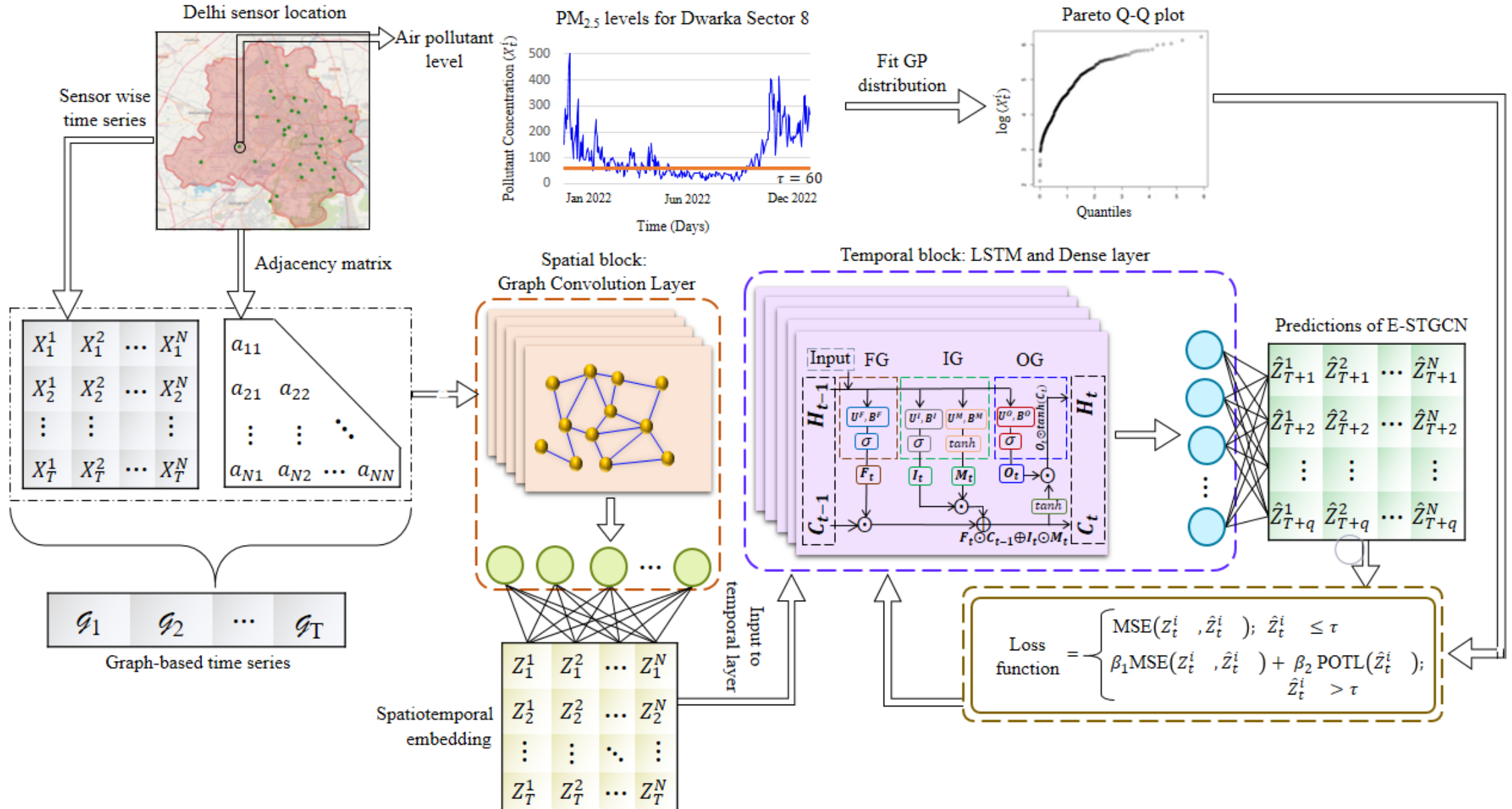
Hyperparameters  
(cross validation)

with

$$\text{PoTL}(z_t^i) = \text{NLL}(\text{GPD}) = \log g(\hat{\alpha}^i) + \left(1 + \frac{1}{\hat{\xi}^i}\right) \log \left(1 + \frac{\hat{\xi}^i z_t^i}{\hat{\alpha}^i}\right)$$

denoting the **negative** log-likelihood of the fitted GPD for location  $i$  for one data point. A likelihood function measures how well a statistical model explains observed data

# E-STGCN Architecture



# Performance and Predictions

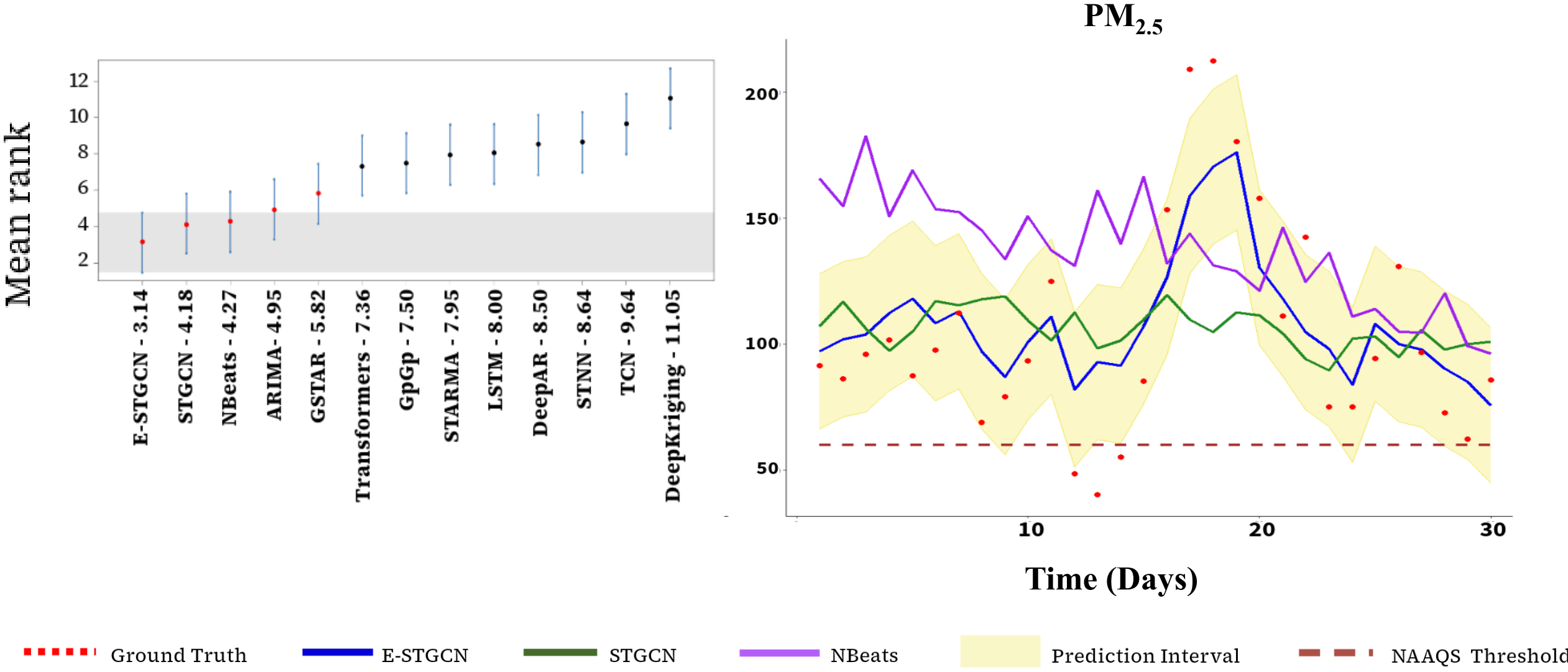


Fig: Ground truth and forecasts of PM<sub>2.5</sub> monitored at Dr. Karni Singh Shooting Range, Delhi during Feb 2023

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